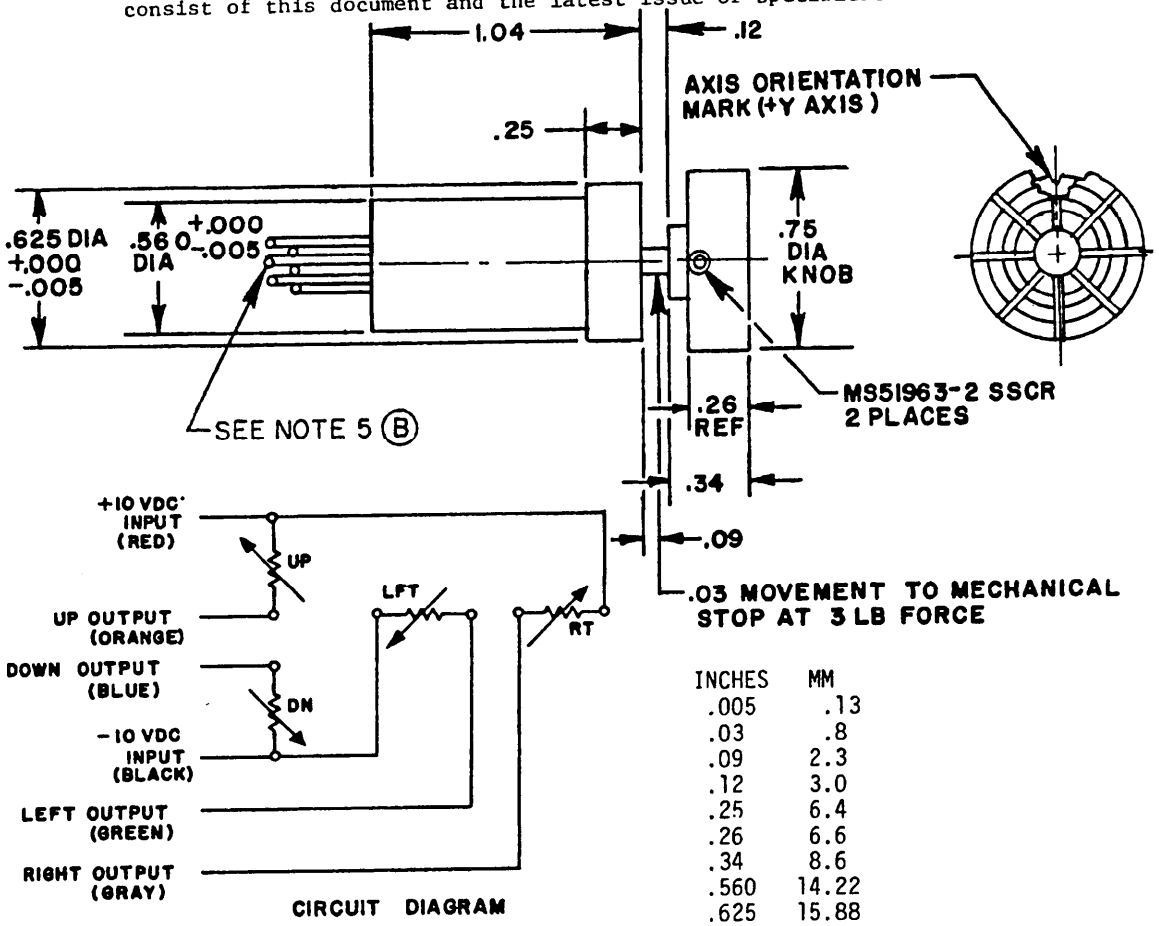
MILITARY SPECIFICATION SHEET

SWITCH, SOLID STATE TRANSDUCER,
2-AXIS JOYSTICK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-81619.



CIRCUIT DIAGRAM

INCHES	MM
.005	.13
.03	.8
.09	2.3
.12	3.0
.25	6.4
.26	6.6
.34	8.6
.560	14.22
.625	15.88
.75	19.0
1.04	26.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Identification placed on body surface.
4. Unless otherwise specified, tolerances are ±.01 inch (.3 mm) on decimals and ±.5° on angles.
5. 6 wire lead terminals shall be a minimum of 36" long and color coded per circuit diagram.

FIGURE 1. Configuration and dimensions.

(B) denotes changes

FSC 5930

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REQUIREMENTS:

OPERATIONS AND DIMENSIONS

Dimensions: See figure.
Operating force: 3.1 pounds maximum (output proportional to force at a rate of .45 ±20% per pound of force on either axis).
Pretravel: Not applicable.
Movement differential: Not applicable.
Overtravel: Not applicable.
(I) Terminals: Flexible wire (MIL-W-22759/7, P/N M22759/7-22..
Weight: 0.06 pound maximum.
Part number: M81619/08-001.

MECHANICAL

Strength of actuating means: 15.0 pounds each direction along two perpendicular axes.
Strength of mounting means: Not applicable.
Terminal strength: Not applicable.
Resistance to soldering heat: Not applicable.
Solderability: Not applicable.
Classification: Class 2 equipment per MIL-E-5400.
Enclosure type: Type I, unsealed.

ELECTRICAL

Resolution: Infinite.
Null temperature coefficient: ±.04% full scale per degree C.
Null hysteresis: ±.5% full scale within 1 second after release of shaft.
Sensitivity temperature coefficient: ±.2% full scale per degree C.
Resistance matching: 1% between each pair of elements.
(B) Absolute supply voltage: 12 volts dc maximum (load impedance 20,000 ohms ±5%).
Absolute supply current: 30 mA maximum.
Supply voltage: 20 Vdc (±10 Vdc Bi-Polar).
(B) Output voltage and current: .45 Vdc ±20% per pound force.
1.54 V maximum, 20 mA maximum.
Load impedance: Not applicable.
Output: Rise time: Not applicable.
Fall time: Not applicable.
Output voltage waveform: Not applicable.
Short circuit: Applicable.
Transient voltage: Applicable.
Insulation resistance: Applicable.
Dielectric withstanding voltage: Applicable.

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ENVIRONMENTAL

Temperature
  Operating: -54°C to +71°C.
  Storage: -54°C to +71°C.
Shock
  Mechanical: 30 g for 11 milliseconds (no inspection during test).
  Temperature: -54°C and +71°C.
Vibration: Applicable (no inspection during test).
Acceleration: 10 g (no inspection during test).
Salt fog: Applicable.
Humidity: Applicable.
Sand and dust: Applicable.
Electromagnetic compatibility: Not applicable.
Magnetic compatibility: Not applicable.
Temperature-altitude: Applicable (no inspection during test).
  Altitude: Sea level to 70,000 feet.
Life: 150,000 cycles to maximum travel and return (no inspection during test).
Conditioning: Not applicable.

Custodian:
  Air Force - 11
  Army - EL
  Navy - AS

Review activity:
  Air Force - 85, 99
  Army - MI
  DLA - ES

Preparing activity:
  Air Force - 11

Agent:
  DLA - ES

(Project 5930-1074-4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-S-81619/8B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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